WORKED SOLUTION

STEP II 1997, Question 2

Pure • Recurrence Relations

STEP II 1997, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q2

The key idea

The chain of equalities is really a recurrence in disguise: each link says that $x_n + 2/x_{n+1} = 3$, so each term determines the next. Compute the first few terms as fractions and look hard at the numerators and denominators. Powers of 2 appear quickly, and once we have a guess, the recurrence gives exactly the step an induction needs.

Solution

Finding the pattern

Reading the chain one equality at a time gives

$$\frac{2}{x_1} = 3 \qquad \text{and} \qquad x_n + \frac{2}{x_{n+1}} = 3 \quad (n \geq 1),$$

so that

$$x_1 = \frac{2}{3}, \qquad x_{n+1} = \frac{2}{3 - x_n}.$$

(The recurrence only makes sense if $x_n \neq 3$; we shall see that every x_n lies between 0 and 1.) Working out the first few terms:

$$x_1 = \frac{2}{3}, \qquad x_2 = \frac{2}{3 - \frac{2}{3}} = \frac{6}{7}, \qquad x_3 = \frac{2}{3 - \frac{6}{7}} = \frac{14}{15}, \qquad x_4 = \frac{2}{3 - \frac{14}{15}} = \frac{30}{31}.$$

The denominators $3, 7, 15, 31$ are each one less than a power of 2, namely $2^{n+1} - 1$, and each numerator is one less than its denominator. So we guess

$$x_n = \frac{2^{n+1} - 2}{2^{n+1} - 1} = 1 - \frac{1}{2^{n+1} - 1}.$$

Proof by induction

Let $P(n)$ be the statement $x_n = \dfrac{2^{n+1} - 2}{2^{n+1} - 1}$.

Base case. For $n = 1$ the formula gives $\frac{4-2}{4-1} = \frac{2}{3} = x_1$, so $P(1)$ is true.

Inductive step. Suppose $P(k)$ is true for some $k \geq 1$. To keep the algebra tidy, write $N = 2^{k+1}$, so that $x_k = \dfrac{N-2}{N-1}$. Then

$$3 - x_k = \frac{3(N-1) - (N-2)}{N-1} = \frac{2N-1}{N-1},$$

which is not zero (indeed it is greater than 2, since $x_k < 1$), so we may divide by it:

$$x_{k+1} = \frac{2}{3 - x_k} = \frac{2(N-1)}{2N-1} = \frac{2N-2}{2N-1}.$$

? **Ask yourself** Why is this the formula with $n = k+1$?

Because $2N = 2^{k+2} = 2^{(k+1)+1}$. So

$$x_{k+1} = \frac{2^{(k+1)+1} - 2}{2^{(k+1)+1} - 1},$$

which is exactly $P(k+1)$.

Since $P(1)$ is true and $P(k) \Rightarrow P(k+1)$ for every $k \geq 1$, $P(n)$ is true for all $n \geq 1$, by induction. This also confirms that $0 < x_n < 1$ for every n, so no denominator $3 - x_n$ is ever zero.

An "otherwise" route: linearising

The guess can also be *derived*. A recurrence of the form $x_{n+1} = 2/(3 - x_n)$ becomes linear if we write $x_n = 2a_n/a_{n+1}$ for a new sequence (a_n) with non-zero terms. Substituting,

$$\frac{2a_{n+1}}{a_{n+2}} = \frac{2}{3 - \frac{2a_n}{a_{n+1}}} = \frac{2a_{n+1}}{3a_{n+1} - 2a_n}, \qquad \text{so} \qquad a_{n+2} = 3a_{n+1} - 2a_n.$$

The auxiliary equation $t^2 - 3t + 2 = 0$ has roots 1 and 2, so $a_n = A + B \cdot 2^n$. Taking $a_1 = 1$ (we are free to scale a) and $a_2 = 3$ (so that $x_1 = 2a_1/a_2 = \frac{2}{3}$) gives $a_n = 2^n - 1$, and then $x_n = \dfrac{2(2^n - 1)}{2^{n+1} - 1}$, the same formula.

Checks

- The formula with $n = 0$ gives $x_0 = 0$, and indeed the first equation $3 = 2/x_1$ is the general link $x_0 + 2/x_1 = 3$ with $x_0 = 0$. That's a nice consistency check.
- As $n \to \infty$, $x_n \to 1$. A limit L of the recurrence must satisfy $L = 2/(3 - L)$, that is $L^2 - 3L + 2 = 0$, so $L = 1$ or $L = 2$, and 1 is the one our sequence approaches from below.
- Iterating $x_{n+1} = 2/(3 - x_n)$ in exact fractions from $x_1 = \frac{2}{3}$ matched the formula for every n up to 40; for example $x_5 = \frac{62}{63}$.

Common slips

- Misreading the chain: the equation is $x_n + 2/x_{n+1} = 3$, not $x_{n+1} + 2/x_n = 3$. Getting this wrong gives a different sequence.
- Stopping after "the pattern continues". The question asks for a proof, and a few terms are only evidence.
- In the inductive step, forgetting to rewrite $2 \cdot 2^{k+1}$ as $2^{(k+1)+1}$, so the conclusion is never visibly the statement $P(k+1)$.

Beyond STEP

Recurrences of the form $x_{n+1} = (ax_n + b)/(cx_n + d)$ are iterations of a Möbius transformation, and they can always be linearised as in the "otherwise" route. Equivalently, we can track powers of the 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, whose eigenvalues (here 1 and 2) control the long-term behaviour. The fixed points 1 and 2 are attracting and repelling respectively, which is why the sequence settles at 1. For the matrix viewpoint, see Strang, *Introduction to Linear Algebra*; for the convergence of sequences in general, Abbott, *Understanding Analysis*.

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